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osapiens Terra

Munich, Germany

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Re: Machine Learning Engineer (m/f/x) – Geospatial AI, osapiens Terra

Dear Hiring Team,

I am applying for the Machine Learning Engineer position in your Geospatial AI team. My background is in computer vision and deep learning with a specific focus on segmentation – which is the core technical foundation your deforestation detection and satellite imagery analysis systems are built on.

During my M.Sc. thesis at FAU's Pattern Recognition Lab, I implemented, extended, and benchmarked a family of transformer-based segmentation architectures – Swin-UNet, MISSFormer, and a custom hybrid CNN–Transformer model – for pixel-level scene analysis on challenging real-world datasets. This required designing encoder-decoder architectures for accurate spatial prediction, evaluating models rigorously across IoU, F1, and Dice Loss, and building an end-to-end inference pipeline suited for deployment. In parallel, my academic project involved implementing DeepLabV3+ and UNet++ with multiple backbone networks for outdoor scene segmentation, conducting a systematic comparative study of fine-tuning strategies across more than a dozen model configurations. This type of work – choosing the right architecture, understanding generalisation across diverse geographies and conditions, and building reliable evaluation pipelines – maps directly to improving and extending production ML models for geospatial analytics.

I have not worked with satellite imagery or geospatial-specific libraries such as Rasterio or GDAL, and I want to be upfront about that. However, the perception tasks involved – multi-class spatial segmentation, change detection, working with large image datasets – are directly within my experience. The domain knowledge required is learnable; the deep learning and engineering skills your core requirements describe are ones I have applied in practice.

On the engineering side, I have built production-ready inference APIs using Detectron2 and FastAPI, containerised with Docker, automated with GitHub Actions CI/CD, and deployed on cloud infrastructure. I am comfortable working across the full ML lifecycle from experimentation through deployment and monitoring.

The mission of osapiens Terra – applying ML to environmental risk detection at scale – is the kind of work I want to build my career on. I would welcome the opportunity to discuss how my background fits the team's needs.

Yours sincerely,

Nitesh Kumar Shah